

DSAIL- HW13

Tomas Salaz

May 7, 2026

Problem 1

To implement an Extensible Array with the operations: **AddToFront**, **AddToEnd**, and **Lookup**, we can maintain 2 Dynamic Arrays, **Front[]** and **Back[]**, where **Front[]** is stored in reverse.

Let $f = \text{len}(\text{front})$ and $b = \text{len}(\text{back})$ thus the total length of the array is $n = f + b$

When Constructing the 2 Extensible Array let **Front[]** contain the first f elements and **Back[]** contain the remaining b elements, thus **Front** = $[a_f, a_{f-1}, \dots, a_2, a_1]$ and **Back** = $[a_{f+1}, a_{f+2}, \dots, a_{n-1}, a_n]$

The Operations **AddToFront** and **AddToEnd** use the standard Dynamic Array Insertion so they double when full.

The Operation **Lookup** is implemented as follows

$$\text{Lookup}(x) = \begin{cases} \text{NULL} & \text{if } x > f + b \\ \text{Front}[f-1-x] & \text{if } 0 \leq x < f \\ \text{Back}[x-f] & \text{if } f \leq x < f + b \end{cases}$$

The **Runtime** of **Lookup** has amortized time of $O(1)$ since we do a constant number of comparisons and one array access.

For **AddToFront** and **AddToEnd** has amortized time of $O(1)$, we know this from since a series of n operations takes $O(n)$ time which is $O(1)$ cost per operation.

Problem 2

To show that **Sub-Iso** is NP-Complete, we have to show that (1) there is a verifiable solution to **Sub-Iso** that has a polynomial Runtime (for NP) AND that (2) **G** has a **Clique** instance (G, k) IFF G_1 is Isomorphic to subgraph G_2 .
(1) Given a function $F : V(G_1) \rightarrow V(G_2)$, that maps the vertices of G_1 to G_2 , is verifiable and runs in poly time if it holds 2 conditions:

1. F is 1-to-1. One scan of F marks each vertex used in G_2 , if any vertex is used twice then the certificate is False, the runtime of this $O(|V(G_1)|)$, which is poly.

2. F preserves edges. For every edge $(u, v) \in E(G_1)$, check that $(F(u), F(v)) \in E(G_2)$ using an Adjacency Matrix for G_2 , the runtime of this is $O(E|G_1|)$ which is poly.

Thus the total time to check the certificate is $O(|V(G_1)| + |E(G_1)|)$, which is poly thus is NP.

(2) For the Reduction from **Clique** to **Sub-Iso**, we must show that G has a **Clique** instance (G, k) IFF G_1 is Isomorphic to subgraph G_2 .

The Construction for this reduction is given the inputs of **Clique**, (G, k) we map to **Sub-Iso** inputs (G_1, G_2) where $G_1 \rightarrow K$, where K is the clique in G AND $G_2 \rightarrow G$, where G is the original graph. This construction take $O(k^2)$ time since we only need to build the Adjacency Matrix for the Clique.

The Forward Direction assumes that G has a clique $C \subseteq V(G)$ where $|C| = k$. By definition, a every pair in a of vertices in a clique C is connected by an edge. Consider the subgraph G_c in G , that comes from the clique C . It has k vertices and contains every edge in between the pairs of vertices. This is the exact same construction of K , so $G_c = K$ this is isomorphic to G .

The Backwards Direction assumes that there is K is isomorphic in G by the mapping $F : V(K) \rightarrow V(G)$. Let $C = F(V(K)) \subseteq V(G)$. We must maintain the 2 conditions:

- $|C| = k$. F is isomorphic since it is 1-to-1, so the k vertices in K map to k distinct vertices in G .
- C is a Clique so for any distinct vertices, $(u, v) \in V(K)$. Since F preserves edges $(F(u), F(v)) \in E(G)$, thus the edges are preserved.

Thus C is a clique in G

Therefore **Sub-Iso** is NP-Complete.

Problem 3

Weighted-Item-Cover (WIC) can be shown to be NP-Hard by using a reduction from **Vertex-Cover**.

Given a **Vtx-Cov** instance (G, k) , where $G = (V, E)$, we can construct an instance of **WIC** with inputs (S, T, W) .

We have to first map $S \rightarrow V$, so each item is a vertex in G .

Then we map $T = \{u, v\} : (u, v) \in E$, so each subset in T gets an edge in G .

The last step is to map $W \rightarrow k$ and set each weight to 1, so $w(G(V)) = 1$. By setting each weight to 1, the constraint, *total weight* $\leq W$ in **WIC** maps to the *number of items* $\leq k$ constraint in **Vtx-Cov**.

This construction take $O(|V| + |E|)$ time which is polynomial.

For the **Proof fo Correctness**: we must show that G has a **Vertex Cover** of size k IFF there exists $S' \subseteq S$ where $w(S') \leq W$ and S' covers every subset in T .

Forward Direction: Assume G has a **Vtx-Cov** $V' \subseteq V$, where $|V'| \leq k$. To treat the vertices as items, Let $S' = V'$. The **Weight Constraint** holds here since each item has weight 1, $w(S'(V)) = 1$. The total weight of S' is $|S'| = |V'| \leq k = W$, thus the constraint is fulfilled. The **Covering Constraint** says that for any subset $\{u, v\} \in T$. By the construction of **WIC**, the subset $\{u, v\} \in T$ corresponds to an edge $(u, v) \in E$. Since V' is a **Vtx-Cov**, at least u or v is in V' thus is in S' , so the subset fulfills the Covering Constraint.

Backwards Direction: Assuming $S' \subseteq S$, where $w(S') \leq W$ and covers every subset in T . Let $V' = S'$, so items are vertices in G . The **Size Constraint** holds since each item has weight 1, thus $|V'| = |S'| = w(S') \leq W = k$. The **Cover Constraint** is also held, since we can take any edge $(u, v) \in E$. By Construction, $\{u, v\} \in T$ and since S' covers this subset at least u or v is in $S' = V'$, thus the edge $(u, v) \in V'$ do it is covered.

Problem 4

Part A: Reduction from Clique

Given an instance of **Clique**, we have the inputs (G, k) we can map to an instance of **Max-Inside-Edges** (MIE) which has inputs (G, k) . We can map $G \rightarrow G$ and $k \rightarrow x$. To Solve the problem, we call MIS on (G, k) to get the maximum number of inside edges, called M . Then we have a subroutine, in which we return TRUE to **Clique** IFF $M = \binom{k}{2}$

This construction runs in poly time since we are only creating a Graph.

Proof Of Correctness: G has a **Clique** C of size k IFF the Max number of Inside Edges over all subsets $S \subseteq V$ where $|S| \leq k$, this is exactly $\binom{k}{2}$

Forward Direction: Assuming G has a clique C of size k , Let $S = C$ so that $|S| = k$. Since every pair of vertices in C are connected by edge, the Number of Inside Edges is $\binom{k}{2}$, which is at least the max. Since the Min cannot exceed $\binom{k}{2}$, any set S with $|S| \leq k$ has at most $\binom{|S|}{2} \leq \binom{k}{2}$ pairs of vertices hence at most $\binom{k}{2}$ inside edges.

Backward Directions: Assume $M = \binom{k}{2}$. Then there exists some S with $|S| \leq k$ which fulfills $\binom{k}{2}$ inside edges. Since $\binom{|S|}{2} \leq \binom{k}{2}$ requires that $|S| \leq k$ combined with $|S| \leq k$, we get $|S| = k$. Since S fulfills the max possible inside edges, $\binom{k}{2}$, on k vertices, every pair of vertices in S must be connected by an edge, so S is a Clique in G of size k .

Thus **Max-Inside-Edges** is NP-Hard.

Part B: We pick S UAR from all subsets V of size x , then we compute $E[\text{num inside edges}]$

For each edge $e = (u, v) \in E$, we use the Indicator Random Variable

$$X_e = \begin{cases} 1 & \text{if } u \in S \text{ and } v \in S \\ 0 & \text{otherwise} \end{cases}$$

Let $X = \sum_{e \in E} X_e$, be the total inside edges, then by LOE

$$E[X] = \sum_{e \in E} E[X_e] = \sum_{e \in E} Pr(X_e = 1)$$

When computing $Pr(X_e = 1)$ for an edge $e = (u, v)$, this is the probability that both u and v are in S , Where S is UAR from all $\binom{n}{x}$ subsets of size x . The number of subsets of size x that have both u and v is the same as the ways we can choose the remaining $x - 2$ vertices from the other remaining $n - 2$ vertices so

$$Pr[X_e = 1] = \binom{n-2}{x-2} / \binom{n}{x} = \frac{x(x-1)}{n(n-1)}$$

Since the probability is the same for every edge and there are m edges the expectation is

$$E[X] = m \cdot \frac{x(x-1)}{n(n-1)}$$